



**UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION**

COURSE	:	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	:	ICT450/ITS432
EXAMINATION	:	FEBRUARY 2023
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of five (5) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
 - i) the Question Paper
 - ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

QUESTION 1

- a) File-based system is considered outdated after database management systems (DBMS) came into the scene. However, there are still companies which currently use the file-based system such as patients record file in certain clinics in Malaysia. Give **TWO (2)** reasons why file-based system is still being used up until today.

(4 marks)

- b) "Relationships are optional in ERD. Some tables may not assigned to any tables".

Explain the truth behind the statement above. Illustrate your answer.

(4 marks)

- c) Personal computer spreadsheet programs such as Microsoft Excel (shown in Table 1) are widely used by business users. They used to store data and manipulate the data by using a wide range of functions. A common misuse of spreadsheets is a substitute for a database.

Table 1 : Sample data for a spreadsheet program

	A	B	C	D	E
1	Customer No	Customer IC	First Name	Last Name	e-mail
2	100	2021877889	Abdul	Badri	a.abdul
3	101	2021755266	Anwi	BenBrahim	e.anwi
4	102	2021522447	Zainuddin	Hamid	zek90
5	103	2021369874	Kayrie	Jamlus	j.kayrie
6	104	2021123698	Huan	Ong	4d40ng
7	105	2021339693	Ishak	Sabarul	IS4u
8	106	2021745632	Rita	Hasrul	DOH

- i) Explain **THREE (3)** common problems that happened when you use a spreadsheet.

(6 marks)

- ii) Identify **FOUR (4)** candidate keys and **ONE (1)** primary key from the table. Give reason why you choose that primary key.

(6 marks)

QUESTION 2

- a) Within a larger information system, the database is subject to a life cycle as well. It is known as Database Life Cycle (DBLC). A properly designed database provides you with access to up-to-date and accurate information. It consists of six phases which are database initial study, database design, implementation & loading, testing & evaluation, operation and maintenance & evolution. Briefly explain **FOUR (4)** activities in the database initial study.

(8 marks)

- b) Describe **FOUR (4)** limitations of mobile database. Give an example for each limitations.

(8 marks)

- c) Table 2 consists of data about staff in Qish Boutique. Data type for StaffID, Salary and DepartmentID is a number. Data type for StaffName is a text. Salary is ranged from 3000-10000. Consider the following data.

Table 2: STAFF

StaffID	StaffName	Salary	DepartmentID
3033	Siti Ziana	11000	10
3035	Fazurah	3000	20
3036	Haqim Ahmad	6000	20
3034	Ahmad Daniel	150a	10

Based on the data, there are certain rows will certainly fail to be inserted into database. Identify the rows and give a reason according to table property, why the data cannot be inserted.

(4 marks)

QUESTION 3

Figure 1 shows the boarding pass for domestic flight in Malaysia. One customer can visit to many destinations and each destination can be visited by many customers.

Figure 1 : Boarding Pass

				ECONOMY CLASS		
Name: Mohammad / Faten Nur Atiqah Ms						
DEPARTURE 13:10 01 Jul 2022		FROM KUL Terminal M KUALA LUMPUR		TO LGK LANGKAWI		ARRIVAL 14:15 01 Jul 2022
Flight MH1436	Boarding 12:10	Booking Ref 6RAL5I	Seat 24F	Gate	Zone D	Class Y
E-Ticket No 2322464811659	Frequent Flyer No	Tier Level		Seq.No 081	SSR	

The structure of boarding is given as the following relational schema.

BOARDING (CustID, CustFirstName, CustLastName, {FlightID, PlaneType, Capacity, DestinationID, DestinationName}, DateDepart, DateArrive, TimeDepart, TimeArrive, Seat)

- a) Define normalization and describe the importance of having normalization in database design.

(3 marks)

- b) Describe **THREE (3)** requirements for good normalized set of tables. Give an example for each requirements.

(6 marks)

- c) Show all the steps to normalize the structure until third normal form (3NF). The primary key(s), partial dependencies and/or transitive dependencies need to be clearly identified.

(11 marks)

QUESTION 4

Ninja Package is a new startup company located in Jasin, Melaka. You are hired as a database developer to create a package delivery database based on the given relational schema.

RIDER (riderID, riderName, startDate, hpNum)

CUSTOMER (icNum, custName, custPhone, custAddress)

DELIVERY (packageID, dateDeliver, riderID, icNum, status)

*Status of delivery is either delivered or unsuccessful.

Write SQL statements based on the following requirements.

a) Create table DELIVERY.

(4 marks)

b) Find total number of customers that are successfully received package(s) in November 2022.

(4 marks)

c) Find total number of unsuccessful delivery for each riders.

(4 marks)

d) List name of riders that make a delivery/deliveries any packages.

(4 marks)

e) Write SQL command to produce the output as follows:

Package ID	Customer Name	Status
p1	Ayu	delivered
p2	Ibrahim	delivered

(4 marks)

QUESTION 5

Chanteq Boutique is a local dress manufacturer. It sells its products to major distributors who distribute through their own stores in major cities and shopping malls. The primary product line is a casual dress for women.

Chanteq Boutique is in the process of designing a database system. The requirements needed are as following:

- Chanteq Boutique manufactures several products. Product id is a unique identifier. Product name is also need to be stored in this system.
- Each product has been assigned to a different stock-keeping unit (SKU). Each SKU belongs to a specific product. SKU number is a unique identifier of SKU. SKU description is also recorded in this table.
- Chanteq Boutique has created several unique colors for its dress. Each color is uniquely identified by a color id. A color description will describe the color options (e.g. red, blue, navy). Each SKU has only one color. Some colors have been assigned to SKU or may never be used.
- Each retailer will have a contract with Chanteq Boutique based on products it will distribute. A retailer will distribute one or more products, and a particular product will be distributed by more than one retailer. Each retailer has a unique id, name and phone number.
- Retailers have stores, which is the place where they sell the skirt and other products. Each retailer has one or more stores. To identify a store in the Chanteq Boutique database for shipping purposes, all customers have agreed to provide their store id and customer id as confirmation. They also need to write their address.

Based on the given information:

- a) Construct an Entity Relationship Diagram (ERD) complete with all the required tables, attributes, relationships as well as the cardinalities using the Crow's Foot notation that reflects all of the given business rules.

(16 marks)

- b) Briefly explain **TWO (2)** examples of reports that can be generated from ERD in (a).

(4 marks)

END OF QUESTION PAPER